

Applying Exponent Laws*Question Bank - Additional 3.05*

Show your work. Circle your final answer.

For questions 1 through 4, rewrite each expression as an equivalent expression with a positive exponent.

1. $\left(-\frac{4}{9}\right)^{-5}$

2. $\left(\frac{9}{10}\right)^{-6}$

3. $\left(\frac{-1}{2}\right)^{-8}$

4. $\left(\frac{2}{7}\right)^{-3}$

Matching.

5. Match each expression on the left with the equivalent expression on the right by writing the letter.

a. $\left(-\frac{13}{9}\right)^{-3}$ A. $\left(\frac{13}{9}\right)^3$ B. $\left(-\frac{13}{9}\right)^3$ C. $\left(-\frac{9}{13}\right)^3$ D. $\left(\frac{9}{13}\right)^3$

b. $\left(\frac{13}{9}\right)^{-3}$

c. $\left(-\frac{9}{13}\right)^{-3}$

d. $\left(\frac{9}{13}\right)^{-3}$

For questions 6 through 9, generate an equivalent expression with one base and one positive exponent. Show the laws you use.

6. $\left(-\frac{5}{4}\right)^{-7} \cdot \left(-\frac{5}{4}\right)^{-3}$

7. $\left(\frac{3}{7}\right)^{-5} \div \left(\frac{3}{7}\right)^{-2}$

8. $\left[\left(\frac{4}{5}\right)^{-2}\right]^3 \cdot \left(\frac{4}{5}\right)^8$

9. $\left(\frac{1.5^4}{1.5^2}\right)^{-1}$

Equivalent or not.

10. Consider the expression $\frac{(\frac{1}{4})^3}{4^{-5}} \cdot 4^4$. For each expression below, decide whether it is equivalent, and justify with a law of exponents.

a. $\frac{4^5 \cdot 4^4}{4^3}$

b. $4^{-8} \cdot 4^4$

c. $\left(\frac{1}{4}\right)^{-2} \cdot 4^4$

d. 4^6

11. Which is the value of $\left(-\frac{3}{10}\right)^{-3}$?

- ☐ A. $-\frac{1000}{27}$
- ☐ B. $\frac{1000}{27}$
- ☐ C. $-\frac{27}{1000}$
- ☐ D. $\frac{27}{1000}$

12. Find the value of n that makes each statement true.

a. $\left(\frac{2}{7}\right)^n = \frac{49}{4}$

b. $\left(\frac{1}{3}\right)^n = 27$